Particle Filter Method for New Zealand Exchange Rate Volatility

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1 Introduction

2 Method

Stochastic volatility modeling relies heavily on Bayesian updating to track latent state variables (Kim et al., 1998).

3 Example

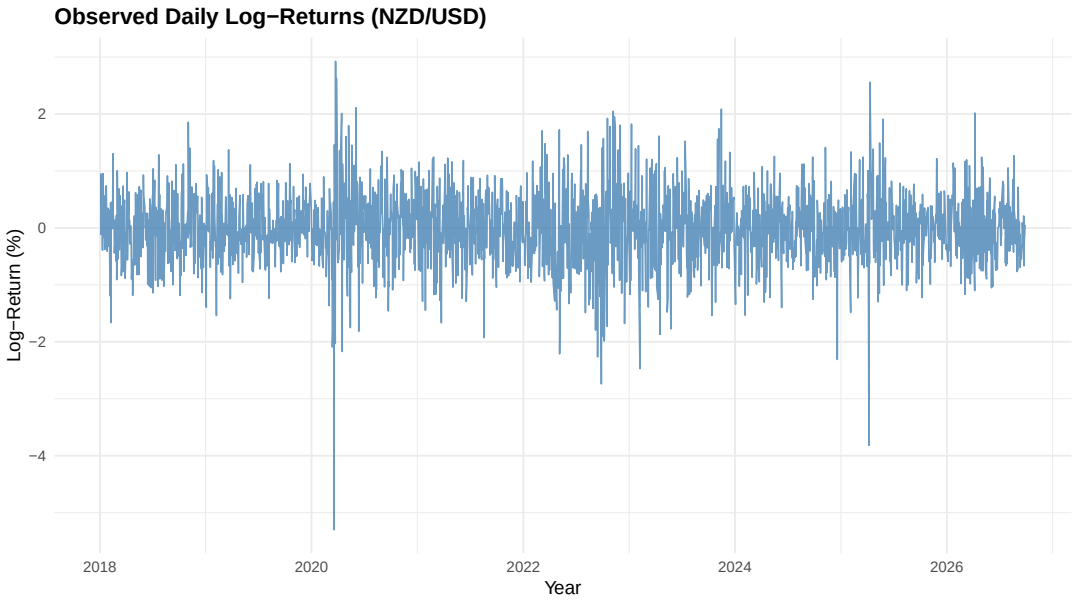


Figure 1: Daily log-returns (y_t) of the NZD/USD exchange rate. The series demonstrates clear volatility clustering, characterized by alternating periods of high market turbulence and relative stability.

4 Discussion

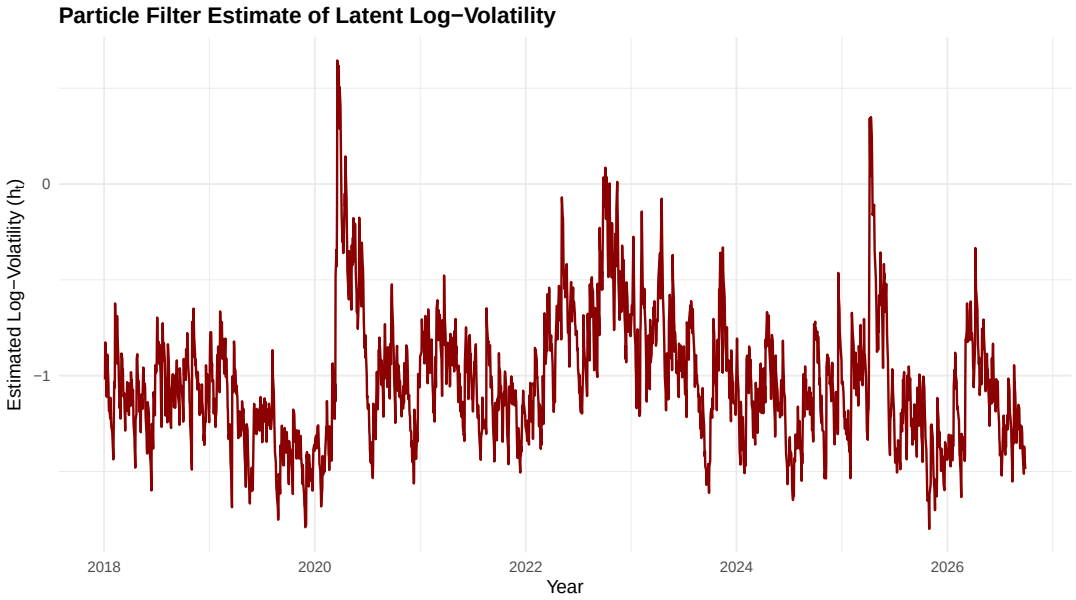


Figure 2: Particle filter estimate of the latent log-volatility state (h_t). The dynamic Bayesian update step enables the filter to continuously track unobserved structural shifts in market variance.

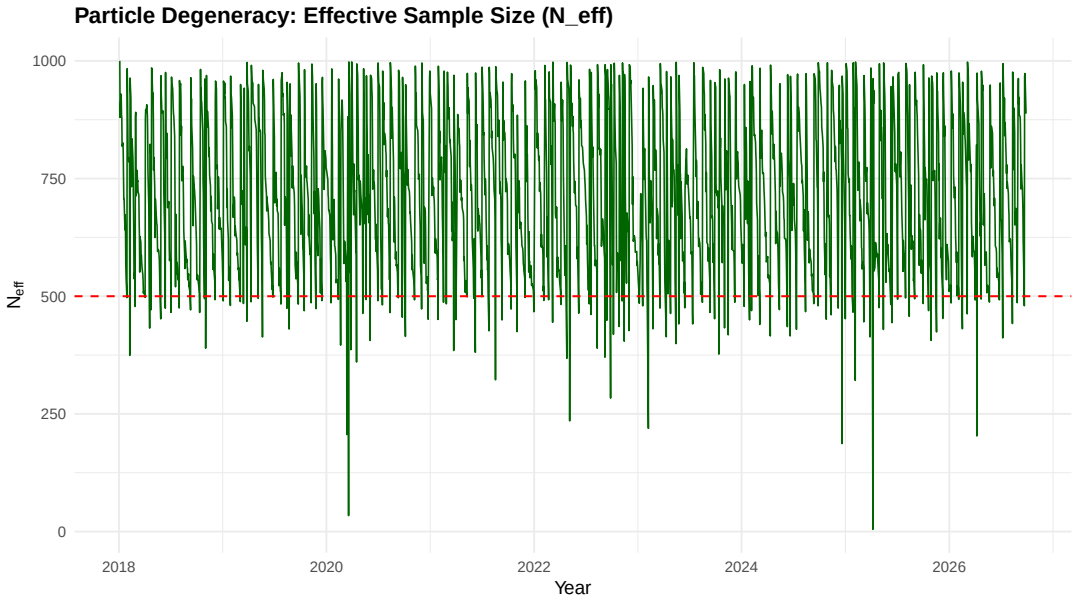


Figure 3: Effective Sample Size (N_{eff}) diagnostic tracking particle degeneracy. The horizontal dashed red line indicates the algorithmic safety net at $M/2 = 500$. The recurrent sawtooth pattern confirms that Sequential Importance Resampling effectively eliminated weak hypotheses and cloned highly weighted particles.

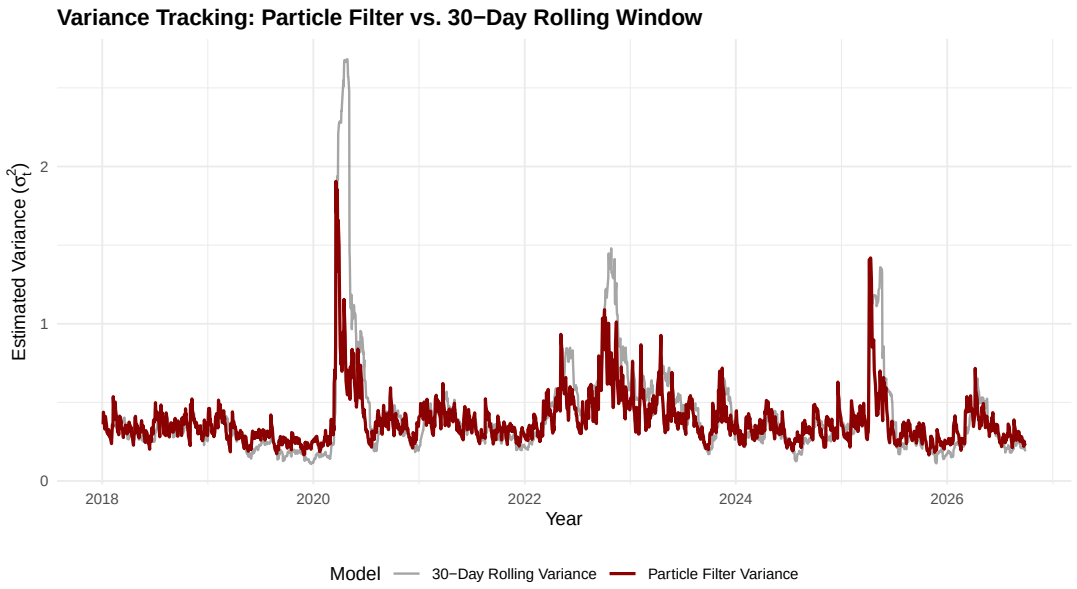


Figure 4: Comparison of estimated variance (σ_t^2) between the stochastic volatility particle filter and a naive 30-day rolling window. The Bayesian likelihood weighting reacts instantly to market shocks, whereas the rolling window demonstrates a rigid structural lag.

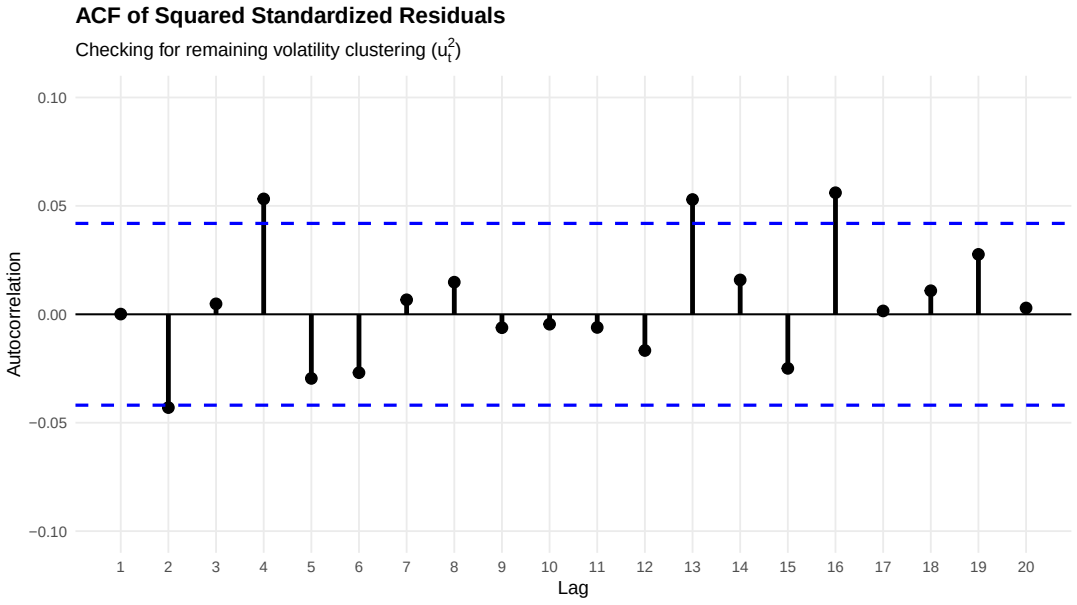


Figure 5: Autocorrelation function (ACF) of the squared standardized residuals (u_t^2). Calibrated with parameters $\phi = 0.93$ and $\sigma_\eta = 0.2$, the lack of significant lags extending beyond the 95% confidence intervals (dashed blue lines) confirms the hidden state successfully absorbed the underlying volatility clustering.

Code Accessibility

Code used for simulations can be found at https://github.com/EnetiTofa/Fiscal-Taylor-Rule.

References

Kim, S., Shephard, N., & Chib, S. (1998). Stochastic volatility: Likelihood inference and comparison with arch models. *The Review of Economic Studies*, *65*(3), 361–393. https://doi.org/10.1111/1467-937X.00050